Adversarial Quiz Explanations

Google Form Link: https://forms.gle/psxVpkW1Uvk92htNA

This document explains why each correct answer is correct and why each incorrect option is a plausible misconception or edge-case trap.

Q1. What does a convolution kernel primarily do?

- A. Sort pixels
- B. Multiply and sum over a region
- C. Normalize image
- D. Compress image

Why B is Correct

A convolution kernel slides across an image region, multiplying kernel values with pixel values and summing the results to extract features. Why the Other Options Are Tempting

- A. Sort pixels
  Sounds computational, but convolutions never rearrange pixel order.
- C. Normalize image
  Normalization is a preprocessing step, not the kernel's main role.
- D. Compress image
  Convolution may reduce dimensions indirectly, but its primary purpose is featuring extraction.

Q2. What is the output size if: N =

32, F = 3, P = 1, S = 1

- A. 30
- B. 31
- C. 32
- D. 33

Formula

Output = ((N − F + 2P) / S) + 1

Output = ((32 − 3 + 2(1)) / 1) + 1
Output = (31 / 1) + 1

Output = 32

Why C is Correct

Padding preserves the original spatial dimensions when using a 3×3 filter and stride 1.

Why the Other Options Are Tempting

- A. 30
  Happens when padding is ignored.
- B. 31
  Common mistake from forgetting the final “+1”.
- D. 33
  Caused by adding padding incorrectly.

Q3. Increasing stride results in:

- A. Larger output
- B. Smaller output
- C. More padding
- D. More channels

Why B is Correct

A larger stride skips more positions while sliding the kernel, reducing output dimensions. Why the

Other Options Are Tempting

- A. Larger output
  Students sometimes assume “bigger stride” means “bigger result.”
- C. More padding
  Padding and stride are independent parameters.
- D. More channels
  Channels are determined by filter count, not stride.

Q4. Which filter is best for noise reduction?

- A. Sobel
- B. Laplacian
- C. Gaussian
- D. Max pooling

Why C is Correct

Gaussian filters smooth images and suppress high-frequency noise.

Why the Other Options Are Tempting

- A. Sobel

Sobel detects edges rather than reducing noise.

- B. Laplacian
  Laplacian often amplifies noise.

- D. Max pooling
  Pooling reduces dimensions but is not a smoothing filter.

Q5. Sobel filter is used for:

- A. Blurring
- B. Edge detection
- C. Compression
- D. Color correction

Why B is Correct

Sobel filters compute image gradients to highlight edges. Why the

Other Options Are Tempting

- A. Blurring
  Gaussian filtering performs blurring.

- C. Compression
  Compression is unrelated to edge filtering.

- D. Color correction
  Sobel works on gradients, not color balancing.

Q6. Padding is used to:

- A. Increase computation
- B. Preserve spatial size
- C. Reduce channels
- D. Normalize input

Why B is Correct

Padding helps maintain output dimensions and preserves edge information. Why the

Other Options Are Tempting

- A. Increase computation
  Padding can slightly increase operations, but that is not its purpose.

- C. Reduce channels
  Channels depend on filters.

- D. Normalize input
  Normalization is separate from padding.

Q7. Depthwise convolution applies:

- A. One filter per channel
- B. One filter to all channels
- C. No filters
- D. Random filters

Why A is Correct

Depthwise convolution processes each channel independently with its own filter.

Why the Other Options Are Tempting

- B. One filter to all channels
  This resembles standard convolution.
- C. No filters
  Convolution always uses filters.
- D. Random filters
  Filters are learned during training.

Q8. Pointwise convolution uses:

- A. 3×3 filters
- B. 5×5 filters
- C. 1×1 filters
- D. No filters

Why C is Correct

Pointwise convolution uses 1×1 filters to combine information across channels. Why the

Other Options Are Tempting

- A. 3×3 filters
  Common convolution size but not pointwise convolution.
- B. 5×5 filters
  Larger spatial filter, not pointwise.
- D. No filters
  Pointwise convolution still uses learnable kernels.

Q9. Laplacian filter detects:

- A. Color
- B. Texture
- C. Noise
- D. Second-order edges

Why D is Correct

The Laplacian operator detects rapid intensity changes using second derivatives. Why the

Other Options Are Tempting

- A. Color
  Laplacian works on intensity changes, not color.
- B. Texture
  Texture analysis is broader and indirect.
- C. Noise
  Laplacian may amplify noise but does not specifically detect it.

Q10. Which parameter increases computation most?

- A. Padding
- B. Kernel size
- C. Stride
- D. Activation

Why B is Correct

The number of multiplications per convolution position scales with K^2. Assuming fixed channel depth, kernel size has the greatest direct impact on computation per position compared to padding, stride, or activation functions.

- A. Padding
  Padding slightly increases the input size and therefore total operations, but the effect is small compared to kernel size growth.
- C. Stride
  Larger stride actually reduces computation by skipping positions.
- D. Activation
  Activation functions add computation but are applied elementwise after convolution and contribute far less than a large kernel.